# Leaf Mass on the Sunny and Shady Sides of a Tree

Your Name

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is a comment in the quarto file 1. Save this file at the TOP of your project folder (next to data/, themes/, figures/, scripts/) and name it leaf_report.qmd 2. Press Preview in the upper left of the editor window right away — it should build before you add anything. 3. The analysis code is already here and runs. Your job: add the finishing touches marked # TODO: (colors, labels, theme, table formatting), write the text sections, and render as you go.

KEEP THIS FILE. For any future analysis, copy it, change the data file in the load-data chunk and the column names, and rewrite the text.

Paths like "data/…" work because Quarto runs the code from the folder this file is saved in — the top of your project. —>

# 1 Introduction

- **Null hypothesis:**
- **Alternate hypothesis:**

# 2 Methods

# 3 Results

## 3.1 Box plot

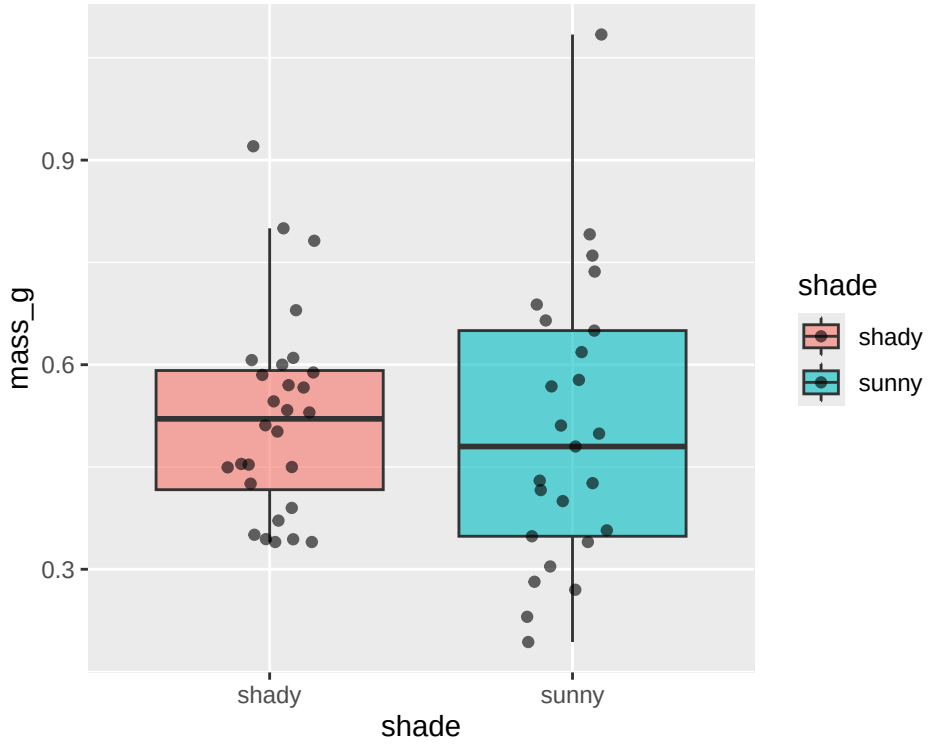


Figure 1: Leaf mass by side; each point is one leaf.

### 3.2 Mean ± SE

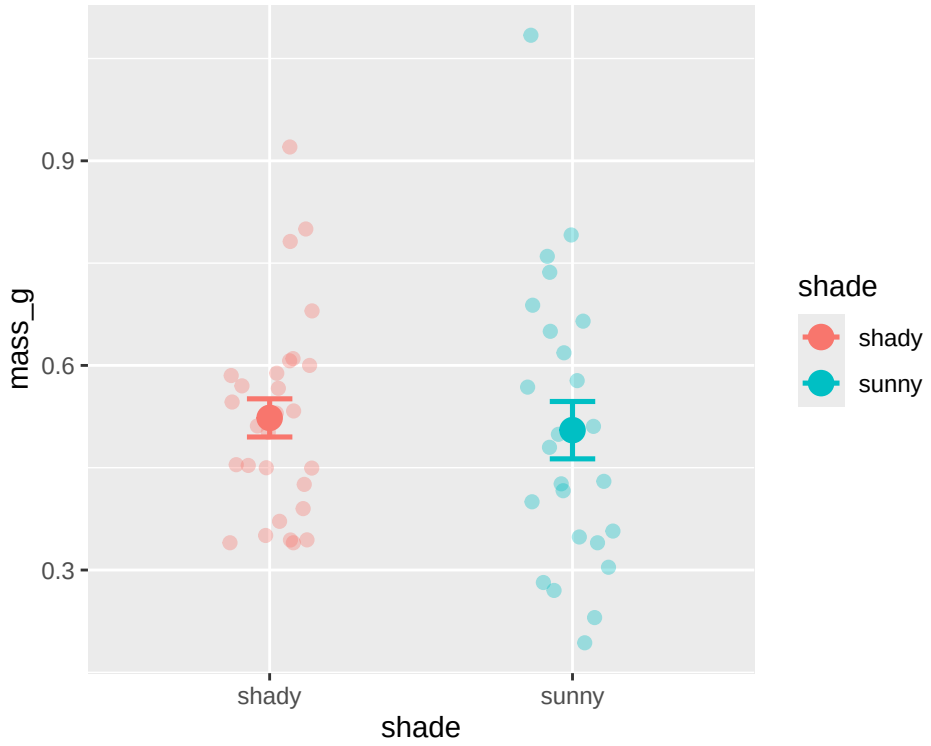


Figure 2: Mean leaf mass (± 1 SE) by side.

## 3.3 Histogram

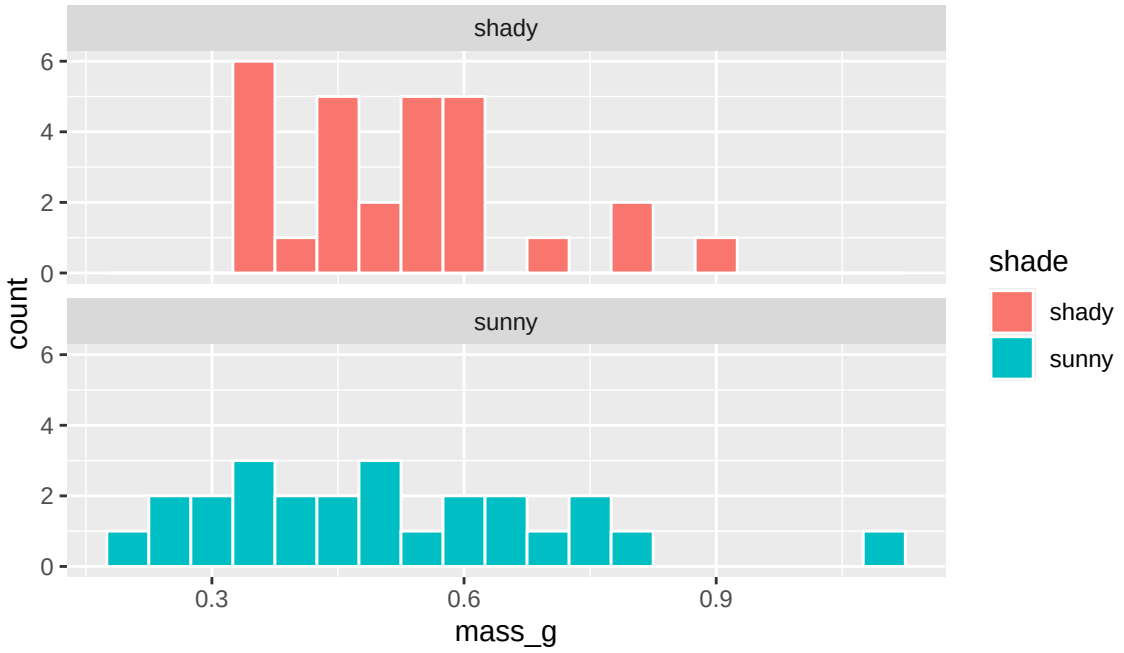


Figure 3: Distribution of leaf mass on each side.

## 3.4 Summary statistics

Table 1: Summary statistics for leaf mass (g) by side.

| shade | n | mean | variance | sd | se | ci_lower | ci_upper |
|---|---|---|---|---|---|---|---|
| shady | 28 | 0.5230357 | 0.0219048 | 0.1480026 | 0.0279699 | 0.4656463 | 0.5804251 |
| sunny | 25 | 0.5050640 | 0.0442281 | 0.2103048 | 0.0420610 | 0.4182544 | 0.5918736 |

## 3.5 Welch's t-test

```
    Welch Two Sample t-test

data:  mass_g by shade
t = 0.35579, df = 42.527, p-value = 0.7238
alternative hypothesis: true difference in means between group shady and group sunny is not e
```

```
95 percent confidence interval:
 -0.08392774  0.11987117
sample estimates:
mean in group shady mean in group sunny
          0.5230357           0.5050640
```

Table 2: Welch's t-test of leaf mass by side of the tree.

| estimate1 | estimate2 | statistic | parameter | p.value | conf.low |
|---|---|---|---|---|---|
| 0.5230357 | 0.505064 | 0.3557927 | 42.52693 | 0.7237547 | -0.0839277 |

# 4 Discussion